

PPS 9.1: Potential Difference, I-V Characteristics, Ohm's Law

1. A filament lamp is described as being 120 V, 60 W. The lamp is connected to a supply so that it lights normally. Which statement is correct?

The lamp transfers 120 joules in 2.0 seconds (C), because the filament lamp is $60\text{W} = 60\text{J s}^{-1}$
Therefore in 2 seconds 120J of energy is transferred.

2. An electron gun is used to accelerate electrons from rest through a voltage V . The electrons emerge with a speed u . The voltage in the gun is halved to $V/2$. At what speed do the electrons emerge?

$$\text{Let } V_2 = \frac{V}{2}.$$

$$\text{We know that } eV = \frac{1}{2}mv^2 \text{ and thus } u = \sqrt{\frac{2eV}{m}} = \sqrt{2}\sqrt{\frac{eV}{m}}.$$

$$\text{Therefore } u_2 = \sqrt{\frac{eV}{m}} = \frac{u}{\sqrt{2}}. \text{ So the speed that the electrons emerge at is } \frac{u}{\sqrt{2}}: \text{ (C)}$$

3. The total energy gained by 20 electrons travelling through a potential difference V is 30 keV. What is the potential difference V ?

$$\text{Therefore the energy gained by 1 electron travelling through a potential difference } V \text{ is } \frac{30000}{20} \text{ eV} = 1500 \text{ V}$$

4. In a particle-accelerator electrons are accelerated through a potential difference of 120 kV. The electron beam current is $8.0 \mu\text{A}$. What is the total energy transferred to the electrons in a time of 2.0 hours?

$$E = VIt = 120000\text{V} \times (8.0 \times 10^{-6})\text{A} \times (2.0 \times 60 \times 60\text{s}) = 6912 \text{ J} \approx 6900 \text{ J (D)}$$

5. The graph shows the I - V characteristic of a semiconductor diode. Which statement about the resistance of the diode can be deduced from the characteristic?

(D) - The resistance decreases between 0.7V and 1.0V, because the gradient gets steeper.

6. A student determines the resistance R of a filament lamp by measuring the potential difference V across it and the current I through it. The values recorded by the student are:

$$V = (5.00 \pm 0.2) \text{ V and } I = (40.0 \pm 0.1) \text{ mA.}$$

What is the percentage uncertainty in the value of R ?

$$U_V = \frac{0.2}{5.00} = 4\%$$

$$U_I = \frac{0.1}{40.0} = 0.25\%$$

$$R = \frac{V}{I} \Rightarrow U_R = U_V + U_I = 4.25\%$$

7.

a.

i. **-1.0V to 2.6V**

Resistance does not change from -1.0V to 2.6V : it is effectively infinite. At -1.0V , the current in mA is 0, and since $R = \frac{V}{I}$, the resistance of the LED is effectively infinite.

ii. **2.6V to 3.0V**

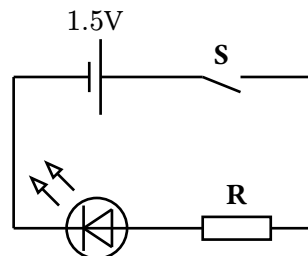
From 2.6V to 3.0V , the resistance decreases sharply. At 2.6V , the resistance is incredibly large ($\text{M}\Omega$), but at 3.0V , the resistance is 430Ω .

iii. **3.0V to 3.4V**

From 3.0V to 3.4V , the resistance decreases at the same rate, but still quickly, going from 430Ω at 3.0V to 120Ω at 3.4V .

- b. The circuit does not work because the LED is pointed in the wrong direction. Current is attempting to flow “backwards” through the LED, so the resistance in the component becomes incredibly high, and no current flows.

The circuit should look like this:



so that the diode is pointing the correct way such that current is able to flow through the circuit.

8.

a. $P = IV; I = \frac{P}{V}; R = \frac{V}{I}$

$$\therefore R = \frac{V^2}{P}$$

$$= \frac{230^2}{3500}$$

$$= 15.114\Omega \approx 15\Omega$$

- b. The heater does not obey Ohm's Law. This is because the resistance in the heater changes as it warms up - the positive ions in the heater vibrate more around their mean position, and the charge carriers travelling through the wire in the heater collide with the positive ions at a greater frequency.

This means that it takes an increasingly large amount of p.d. to move the charge carriers through the wire.

Therefore, resistance increases as the temperature of the heater increases.

9. Component Y was most likely a resistor, which obeys Ohm's law. We know this because the resistance remains constant as voltage and current increase: $V \propto I$.

Component X may have been a filament lamp. This is because the resistance of component X increases as the voltage increases, suggesting an increase in temperature as the voltage increases. This behavior is also observed in a filament lamp.

Extension Questions

Exponential I-V

a) $I = a(e^{bV} - 1)$

Therefore $[A] = [a]^{[b][V]}$

Since you cannot have a unit to the power of another unit because

$$e^x = \sum_{k=0}^{\infty} \frac{x^n}{n!}$$

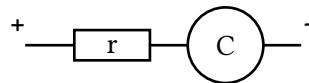
and thus the value of n diverges to infinity, the units of b must be V^{-1} . Therefore, the units of a must be amperes (A).

b)

$$I = a(e^{bV} - 1)$$

$$\begin{aligned} b &= \frac{\ln\left(\frac{I}{a} + 1\right)}{V} \\ &= \frac{\ln\left(\frac{1.0\text{A}}{10^{-4}\text{A}} + 1\right)}{0.23\text{V}} \\ &= 40 \text{ V}^{-1} \end{aligned}$$

c) Let this diagram be the part of the circuit described, where C is the component and r is the cell.



Then the voltage across the non-Ohmic component is

$$\begin{aligned} V &= \frac{\ln\left(\frac{I}{a} + 1\right)}{b} \\ &= \frac{\ln\left(\frac{0.1}{10^{-4}} + 1\right)}{40} \\ &= 0.17\text{V} \end{aligned}$$

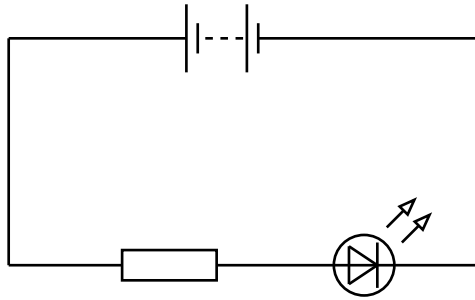
Therefore the voltage lost in the internal resistor of the cell $= 1.5\text{V} - 0.17\text{V} = 1.33\text{V}$.

d) $r = \frac{V}{I} = \frac{1.33\text{V}}{0.1\text{A}} = 13\Omega$

The component is a diode.

Resistor for LED Circuit

Let this diagram be the circuit described.



a) $I = 4.0 \text{ mA} = 4 \times 10^{-3} \text{ A}$.

Then V_R = the voltage across the resistor $= 3.0\text{V} - 1.6\text{V} = 1.4\text{V}$.

If the current in the diode is 4.0 mA, then the current in the resistor must also be 4.0 mA.

Therefore the resistance of the resistor is $R_R = \frac{V_R}{I_R} = \frac{1.6\text{V}}{0.004\text{A}} = 400\Omega$

b) The power draw in the LED can be given as $E_D = 1.6\text{V} \times 0.004\text{A} \times t = (6.4 \times 10^{-3})t$

The power draw in the resistor can be given as $E_R = 1.4\text{V} \times 0.004\text{A} \times t = (5.6 \times 10^{-3})t$

Therefore the ratio between the power draw in the LED vs the power draw in the resistor is

$$E_D : E_R = (6.4 \times 10^{-3})t : (5.6 \times 10^{-3})t$$

As a number, the ratio is $\frac{6.4}{5.6} = 1.1$.